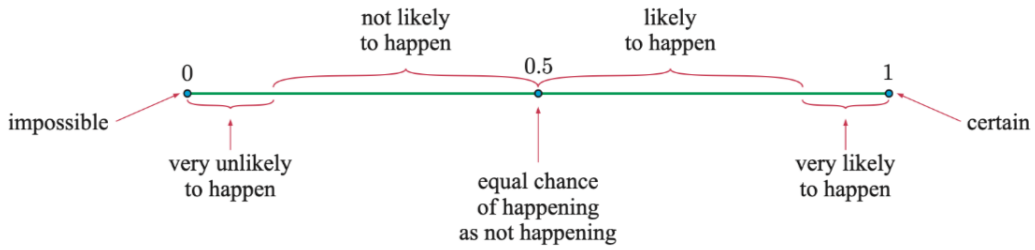
I- Introduction

In the field of probability theory we use mathematics to describe the chance or likelihood of an event happening.



Definition

The number of trials is the total number of times the experiment is repeated.

Definition

The outcomes are the different results possible for one trial of the experiment.

Definition

The frequency of a particular outcome is the number of times that this outcome is observed.

Definition

The relative frequency of an outcome is the frequency of that outcome expressed as a fraction or percentage of the total number of trials.

Example

When a small plastic cone was tossed into the air 279 times it fell on its *side* 183 times and on its *base* 96 times.

We say :

— the number of trials is 279

— the outcomes are *side* and *base*

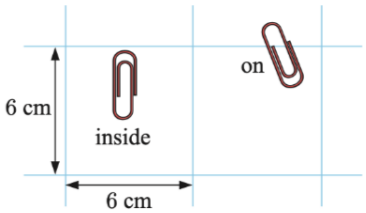
— the frequencies of *side* and *base* are 183 and 96 respectively

— the relative frequencies of *side* and *base* are $\frac{183}{279} \approx 0.656$ and $\frac{96}{279} \approx 0.344$ respectively.

EXERCISE 1

When a batch of 145 paper clips was dropped onto 6 cm by 6 cm squared paper, it was observed that 113 fell completely inside squares and 32 finished up on the grid lines. Find, to 2 decimal places, the experimental probability of a clip falling:

a inside a square b on a line.



EXERCISE 2

Betul records the number of phone calls she receives over a period of consecutive days.

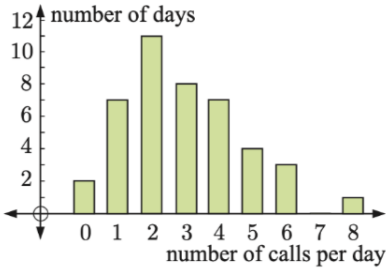
a For how many days did the survey last?

b Estimate Betul's chance of receiving:

i no phone calls in a day

ii 5 or more phone calls in a day

iii less than 3 phone calls in a day.



II- Sample space

a) Definition

Definition

A sample space U is the set of all possible outcomes of an experiment.

It is also referred to as the universal set U.

Example

List the sample space of possible outcomes for :

1. tossing a coin
2. rolling a normal die.

EXERCISE 3

List the sample space for the following:

a twirling a square spinner labelled A, B, C, D

b the sexes of a 2-child family

c the order in which 4 blocks A, B, C and D can be lined up

d the 8 different 3-child families.

b) Representation

There are a variety of ways of representing or illustrating sample spaces, including :

— lists

— tables of outcomes

— 2-dimensional grids

— Venn diagrams

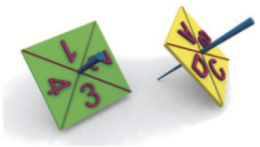
— tree diagrams

Example

Use a 2-dimensional grid to illustrate the possible outcomes when 2 coins are tossed.

EXERCISE 4

Illustrate on a 2-dimensional grid the sample space for:



a rolling a die and tossing a coin simultaneously

b rolling two dice

c rolling a die and spinning a spinner with sides A, B, C, D

d twirling two square spinners, one labelled A, B, C, D and the other 1, 2, 3, 4.

Example

Illustrate, using a tree diagram, the possible outcomes when :

1. tossing two coins
2. drawing two marbles from a bag containing many red, green, and yellow marbles.

EXERCISE 5

Illustrate on a tree diagram the sample space for:

a tossing a 5-cent and a 10-cent coin simultaneously

b tossing a coin and twirling an equilateral triangular spinner labelled A, B, and C

c twirling two equilateral triangular spinners labelled 1, 2, and 3, and X, Y, and Z

d drawing two tickets from a hat containing a large number of pink, blue, and white tickets.

III- Probability rules

a) Equally Likely Outcomes

Property

For an event A containing equally likely possible results, the probability of A occurring is

$$\boxed{\mathrm{P}(A) = \frac{\text{the number of members of the event } A}{\text{the total number of possible outcomes}} = \frac{n(A)}{n(U)}}$$

Example

A ticket is randomly selected from a basket containing 3 green, 4 yellow, and 5 blue tickets. Determine the probability of getting :

a a green ticket

b a green or yellow ticket

c an orange ticket

d a green, yellow, or blue ticket.

Property

For any event A, the probability $\mathrm{P}(A)$ of A occurring satisfies $0 \leq \mathrm{P}(A) \leq 1$.

b) Complementary events

Definition

Two events are complementary if exactly one of the events *must* occur.

Property

If A is an event, then A' is the complementary event of A, or 'not A'.

$$\mathrm{P}(A) + \mathrm{P}(A') = 1$$

Example

An ordinary 6-sided die is rolled once. Determine the chance of :

a getting a 6 **b** not getting a 6

c getting a 1 or 2 **d** not getting a 1 or 2

EXERCISE 6

A marble is randomly selected from a box containing 5 green, 3 red, and 7 blue marbles. Determine the probability that the marble is:

a red **b** green **c** blue

d not red **e** neither green nor blue **f** green or red.

EXERCISE 7

What is the probability that a randomly chosen person has his or her next birthday:

a on a Tuesday **b** on a weekend **c** in July **d** in January or February?

EXERCISE 8

a List the six different orders in which Antti, Kai, and Neda may sit in a row.

b If the three of them sit randomly in a row, determine the probability that:

i Antti sits in the middle **ii** Antti sits at the left end

iii Antti does not sit at the right end **iv** Kai and Neda are seated together.

EXERCISE 9

a List, in systematic order, the 24 different orders in which four people A, B, C, and D may sit in a row.

b Determine the probability that when the four people sit at random in a row:

i A sits on one of the end seats

ii B sits on one of the two middle seats

iii A and B are seated together

iv A, B, and C are seated together, not necessarily in that order.

Example

Use a two-dimensional grid to illustrate the sample space for tossing a coin and rolling a die simultaneously. From this grid determine the probability of :

a tossing a head **b** getting a tail and a 5 **c** getting a tail or a 5.

EXERCISE 10

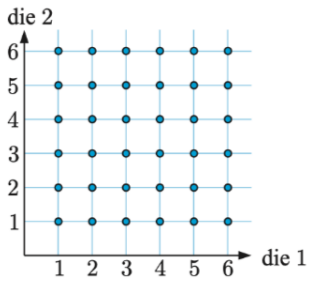
Draw the grid of the sample space when a 5-cent and a 10-cent coin are tossed simultaneously. Hence determine the probability of getting:

a two heads **b** two tails **c** exactly one head **d** at least one head.

EXERCISE 11

A pair of dice is rolled. The 36 different possible results are illustrated in the 2-dimensional grid.
Use the grid to determine the probability of getting:

a two 3s **b** a 5 and a 6
c a 5 or a 6 (or both) **d** at least one 6
e exactly one 6 **f** no sixes
g a sum of 7 **h** a sum greater than 8
i a sum of 7 or 11 **j** a sum of no more than 8.



c) Using table of outcomes

Example

People exiting a new ride at a theme park were asked whether they liked or disliked the ride. The results are shown in the table alongside.

Use this table to estimate the probability that a randomly selected person who went on the ride :

	Child	**Adult**
Liked the ride	55	28
Disliked the ride	17	30

1. liked the ride
2. is a child *and* disliked the ride
3. is an adult *or* disliked the ride
4. liked the ride, given that he or she is a child
5. is an adult, given that he or she disliked the ride.

EXERCISE 12

A sample of adults in a suburb were surveyed about their current employment status and their level of education. The results are summarised in the table below.

	Employed	Unemployed
Attended university	225	34
Did not attend university	197	81

Estimate the probability that the next randomly chosen adult:

a attended university
b did not attend university and is currently employed
c is unemployed
d is employed, given that they attended university
e attended university, given that they are unemployed.

EXERCISE 13

A small hotel in London has kept a record of all the room bookings made for the year, categorised by season and booking type. Find the probability that a randomly selected booking was:

	Single	Double	Family
Peak season	125	220	98
Off-peak season	248	192	152

a in the peak season
b a single room in the off-peak season
c a single or a double room
d a family room, given that it was in the off-peak season
e in the peak season, given that it was not a single room.

IV- Compound events

a) Independent events

Definition

Events are **independent** if the occurrence of either of them does not affect the probability that the others occur.

Property

If A and B are **independent** events then

$$P(A \text{ and } B) = P(A) \times P(B).$$

Example

A coin and a die are tossed simultaneously. Determine the probability of getting a head and a 3 without using a grid.

EXERCISE 14

A coin is tossed 3 times. Determine the probability of getting the following sequences of results:

a head then head then head b tail then head then tail.

EXERCISE 15

A school has two photocopiers. On any one day, machine A has an 8% chance of malfunctioning and machine B has a 12% chance of malfunctioning. Determine the probability that on any one day both machines will:

a malfunction b work effectively.

EXERCISE 16

A couple would like 4 children, none of whom will be adopted. They will be disappointed if the children are not born in the order boy, girl, boy, girl. Determine the probability that they will be:

a happy with the order of arrival b unhappy with the order of arrival.

EXERCISE 17

An archer always hits a circular target with each arrow fired, and hits the bullseye on average 2 out of every 5 shots. If 3 arrows are fired at the target, determine the probability that the bullseye is hit:

a every time

b the first two times, but not on the third shot

c on no occasion.

b) Dependent events

Definition

Two or more events are **dependent** if they are **not independent**.

Events are **dependent** if the occurrence of one of the events *does affect* the occurrence of the other event.

Property

If A and B are dependent events then

$$\mathrm{P}(A \text{ then } B) = \mathrm{P}(A) \times \mathrm{P}(B \text{ given that } A \text{ has occurred}).$$

Example

A box contains 4 red and 2 yellow tickets. Two tickets are randomly selected from the box one by one *without replacement*. Find the probability that :

a both are red

b the first is red and the second is yellow.

Example

A hat contains 20 tickets numbered $1, 2, 3, \ldots, 20$. If 3 tickets are drawn from the hat without replacement, determine the probability that they are all prime numbers.

EXERCISE 18

A bin contains 12 identically shaped chocolates of which 8 are strawberry creams. If 3 chocolates are selected simultaneously from the bin, determine the probability that:

a they are all strawberry creams

b none of them are strawberry creams.

EXERCISE 19

A lottery has 100 tickets which are placed in a barrel. Three tickets are drawn at random from the barrel, without replacement, to decide 3 prizes. If John has 3 tickets in the lottery, determine his probability of winning:

a first prize b first and second prize c all 3 prizes d none of the prizes.

EXERCISE 20

A hat contains 7 names of players in a tennis squad including the captain and the vice captain. If a team of 3 is chosen at random by drawing the names from the hat, determine the probability that it does *not* contain:

a the captain b the captain or the vice captain.

EXERCISE 21

Two students are chosen at random from a group of two girls and five boys, all of different ages. Find the probability that the two students chosen will be:

a two boys b the eldest two students.

c) Tree diagrams

Example

Carl is not having much luck lately. His car will only start 80% of the time and his motorbike will only start 60% of the time.

a. Draw a tree diagram to illustrate this situation.

b. Use the tree diagram to determine the chance that :

i. both will start

ii. Carl can only use his car.

Property

If there is more than one outcome that results in an event occurring, then we need to add the probabilities of these outcomes.

Example

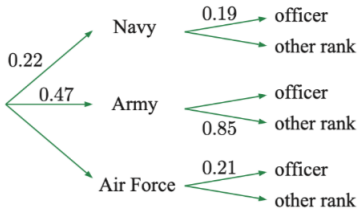
Two boxes each contain 6 petunia plants that are not yet flowering. Box A contains 2 plants that will have purple flowers and 4 plants that will have white flowers. Box B contains 5 plants that will have purple flowers and 1 plant that will have white flowers. A box is selected by tossing a coin, and one plant is removed at random from it. Determine the probability that it will have purple flowers.

EXERCISE 22

a Copy and complete this tree diagram about people in the armed forces.

b What is the probability that a member of the armed forces:

- **i** is an officer
- **ii** is not an officer in the navy
- **iii** is not an army or air force officer?



EXERCISE 23

The probability of rain tomorrow is estimated to be $\frac{1}{5}$. If it does rain, Mudlark will start favourite in the horse race, with probability $\frac{1}{2}$ of winning. If it is fine, he only has a 1 in 20 chance of winning. Display the sample space of possible results of the horse race on a tree diagram. Hence determine the probability that Mudlark will win tomorrow.

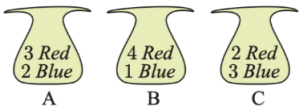
EXERCISE 24

The English Premier League consists of 20 teams. Tottenham is currently in 8th place on the table. It has 20% chance of winning and 50% chance of losing against any team placed above it. If a team is placed below it, Tottenham has a 50% chance of winning and a 30% chance of losing. Find the probability that Tottenham will draw its next game.

EXERCISE 25

Three bags contain different numbers of blue and red marbles.
A bag is selected using a die which has three A faces, two B faces, and one C face. One marble is then selected randomly from the bag.
Determine the probability that the marble is:

a blue **b** red.



d) Sampling

Definition

Suppose we have a large group of objects. If we select one of the objects at random and inspect it for particular features, then this process is known as sampling.

If the object is put back in the group before an object is chosen again, we call it sampling with replacement.

If the object is put to one side, we call it sampling without replacement.

Remark. *Sampling is commonly used in the quality control of industrial processes.*

Example

A box contains 3 red, 2 blue and 1 yellow marble. Find the probability of getting two different colours :

a. if replacement occurs
b. if replacement does not occur.

Example

A bag contains 5 red and 3 blue marbles. Two marbles are drawn simultaneously from the bag. Determine the probability that at least one is red.

EXERCISE 26

5 tickets numbered 1, 2, 3, 4, and 5 are placed in a bag. Two are taken from the bag without replacement.

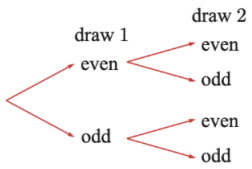
a Complete the tree diagram by writing the probabilities on the branches.

b Determine the probability that:

i both are odd

ii both are even

iii one is odd and the other even.



EXERCISE 27

Marie has a bag of sweets which are all identical in shape. The bag contains 6 orange drops and 4 lemon drops. She selects one sweet at random, eats it, and then takes another at random.

a Determine the probability that:

i both sweets are orange drops

ii both sweets are lemon drops

iii the first is an orange drop and the second is a lemon drop

iv the first is a lemon drop and the second is an orange drop.

b Add your answers in a. Explain why the total must be 1.

EXERCISE 28

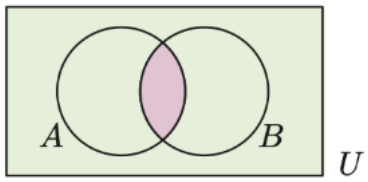
Bag A contains 3 red and 2 white marbles. Bag B contains 4 red and 3 white marbles. One marble is randomly selected from A and its colour noted. If it is red, 2 reds are added to bag B. If it is white, 2 whites are added to bag B. A marble is then selected from bag B. What is the chance that the marble selected is white?

V- Using set notations

a) Venn diagrams

Definition

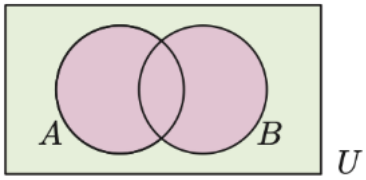
$A \cap B$ denotes the **intersection** of sets A and B. This set contains all elements common to **both** sets.



$$A \cap B = \{x \mid x \in A \textbf{ and } x \in B\}$$

Definition

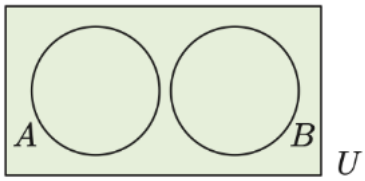
$A \cup B$ denotes the **union** of sets A and B. This set contains all elements belonging to A or B or both A and B.



$$A \cup B = \{x \mid x \in A \textbf{ or } x \in B \textbf{ or } x \in A \cap B\}$$

Definition

Disjoint sets are sets which do not have elements in common.



$$A \cap B = \varnothing$$

where $\varnothing$ denotes the **empty set**. A and B are said to be **mutually exclusive**.

Example

Suppose A is the set of all factors of 24 and B is the set of all factors of 28.

$$U = \{x \mid 1 \leq x \leq 28,\ x \in \mathbb{Z}\}$$

a. Find $A \cap B$.

b. Illustrate A and B on a Venn diagram.

c. Find $A \cup B$.

Example

On separate Venn diagrams containing two events A and B that intersect, shade the region representing :

a. in A but not in B

b. neither in A nor B.

EXERCISE 29

Let A be the set of all factors of 6, B be the set of all positive even integers < 11, and $U = \{x \mid 1 \leqslant x \leqslant 10,\ x \in \mathbb{Z}\}$

a Describe A and B using set notation.

b Illustrate A and B on a Venn diagram.

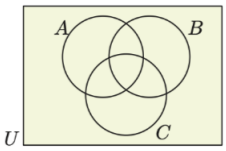
c Find: **i** $n(A)$ **ii** $A \cup B$ **iii** $A \cap B$.

EXERCISE 30

The diagram alongside is the most general case for three events in the same sample space U.
On separate Venn diagram sketches, shade:

a A **b** B' **c** $B \cap C$

d $A \cup C$ **e** $A \cap B \cap C$ **f** $(A \cup B) \cap C$



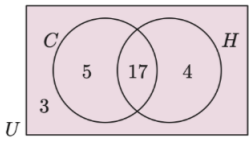
EXERCISE 31

The Venn diagram alongside illustrates the number of students in a particular class who study Chemistry (C) and History (H). Determine the number of students:

a in the class **b** who study both subjects

c who study at least one of the subjects

d who only study Chemistry.



Example

In a class of 30 students, 19 study Physics, 17 study Chemistry, and 15 study both of these subjects. Display this information on a Venn diagram and hence determine the probability that a randomly selected class member studies :

a. both subjects

b. at least one of the subjects

c. Physics but not Chemistry

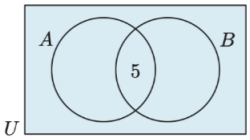
d. exactly one of the subjects

e. neither subject

EXERCISE 32

For two events A and B we are given that $n(A \cap B) = 5$, $n(A) = 11$, $n(A \cup B) = 12$, and $n(B') = 8$.

a Complete the Venn diagram:



b Hence find:

i $P(A \cup B)$ **ii** $P(A')$

EXERCISE 33

50 married men were asked whether they gave their wife flowers or chocolates for her last birthday. The results were: 31 gave chocolates, 12 gave flowers, and 5 gave both chocolates and flowers. If one of the married men was chosen at random, determine the probability that he gave his wife:

a chocolates or flowers **b** chocolates but not flowers

c neither chocolates nor flowers.

EXERCISE 34

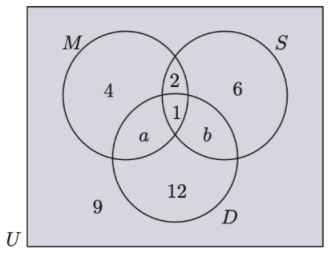
The medical records for a class of 30 children showed that 24 had previously had measles, 12 had previously had measles and mumps, and 26 had previously had at least one of measles or mumps. If one child from the class is selected at random, determine the probability that he or she has had:

a mumps **b** mumps but not measles **c** neither mumps nor measles.

EXERCISE 35

The Venn diagram opposite indicates the types of program a group of 40 individuals watched on television last night.

M represents movies, S represents sport, and D represents drama.



a Given that 10 people watched a movie last night, calculate a and b.

b Find the probability that one of these individuals, selected at random, watched:

i sport **ii** drama and sport **iii** a movie but not sport

iv drama but not a movie **v** drama or a movie.

Example

Verify that

$$(A \cup B)' = A' \cap B'.$$

EXERCISE 36

By drawing appropriate Venn diagrams, verify that:

a $(A \cap B)' = A' \cup B'$ b $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

c $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

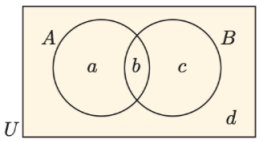
EXERCISE 37

Suppose $S = \{x \mid x \text{ is a positive integer} < 100\}$.
Let $A = \{\text{multiples of 7 in } S\}$ and $B = \{\text{multiples of 5 in } S\}$.

a How many elements are there in: i A ii B iii $A \cap B$ iv $A \cup B$?

b If $n(E)$ represents the number of elements in set E, verify that $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.

c Use the figure alongside to establish that $n(A \cup B) = n(A) + n(B) - n(A \cap B)$ for all sets A and B in a universal set U.



b) Laws of probability

Property

Addition Law :

For two events A and B,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

which means :

$$P(\text{either } A \text{ or } B \text{ or both}) = P(A) + P(B) - P(\text{both } A \text{ and } B).$$

Example

If $P(A) = 0.6$, $P(A \cup B) = 0.7$, and $P(A \cap B) = 0.3$, find $P(B)$.

Property

For disjoints events :

If A and B are **mutually exclusive** events then $P(A \cap B) = 0$

and so the addition law becomes $P(A \cup B) = P(A) + P(B)$.

Example

A chocolate is randomly selected from a box which contains 6 chocolates with hard centres and 12 chocolates with soft centres.

Let H be the event that a randomly selected chocolate from the box has a hard centre, and S be the event that a randomly selected chocolate from the box has a soft centre.

a. Are the events H and S mutually exclusive ?

b. Find : i $P(H)$ ii $P(S)$ iii $P(H \cap S)$ iv $P(H \cup S)$.

Definition

Given two events A and B, the **conditional probability** of A given B is the probability that A occurs given that B has already occurred.

The conditional probability is written $A \mid B$ and read as "A given B".

Example

In a class of 25 students, 14 like pizza and 16 like iced coffee. One student likes neither and 6 students like both. One student is randomly selected from the class. What is the probability that the student :

a. likes pizza

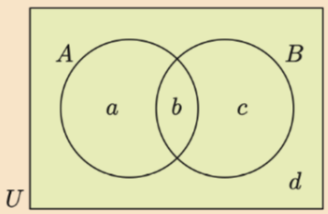
b. likes pizza given that he or she likes iced coffee ?

Property

If A and B are events then

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Proof



$$\mathrm{P}(A \mid B) = \frac{b}{b+c} \quad \{\text{Venn diagram}\}$$

$$= \frac{\frac{b}{(a+b+c+d)}}{\frac{b+c}{(a+b+c+d)}}$$

$$= \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)}$$

Property

$$P(A \cap B) = P(A \mid B)\, P(B)$$

$$P(A \cap B) = P(B \mid A)\, P(A).$$

EXERCISE 38 If $P(A) = 0.4$, $P(A \cup B) = 0.9$, and $P(A \cap B) = 0.1$, find $P(B)$.

EXERCISE 39 If $P(X) = 0.6$, $P(Y) = 0.5$, and $P(X \cup Y) = 0.9$, find $P(X \cap Y)$.

EXERCISE 40 A and B are mutually exclusive events.

If $P(B) = 0.45$ and $P(A \cup B) = 0.8$, find $P(A)$.

Example

In a class of 40 students, 34 like bananas, 22 like pineapple, and 2 dislike both fruits. A student is randomly selected. Find the probability that the student :

a. likes both fruits

b. likes at least one fruit

c. likes bananas given that he or she likes pineapple

d. dislikes pineapple given that he or she likes bananas.

Example

The top shelf in a cupboard contains 3 cans of pumpkin soup and 2 cans of chicken soup. The bottom shelf contains 4 cans of pumpkin soup and 1 can of chicken soup.
Lukas is twice as likely to take a can from the bottom shelf as he is from the top shelf. Suppose Lukas takes one can of soup without looking at the label. Determine the probability that it :

a. is chicken
b. was taken from top shelf given that it is chicken.

EXERCISE 41 In a group of 50 students, 40 study Mathematics, 32 study Physics, and each student studies at least one of these subjects.

1. Use a Venn diagram to find how many students study both subjects.
2. If a student from this group is randomly selected, find the probability that he or she :
   (a) studies Mathematics but not Physics
   (b) studies Physics given that he or she studies Mathematics.

EXERCISE 42 In a group of 40 boys, 23 have dark hair, 18 have brown eyes, and 26 have dark hair, brown eyes, or both. One of the boys is selected at random. Determine the probability that he has :

1. dark hair and brown eyes
2. neither dark hair nor brown eyes
3. dark hair but not brown eyes
4. brown eyes given that he has dark hair.

EXERCISE 43 In a certain town three newspapers are published. 20% of the population read A, 16% read B, 14% read C, 8% read A and B, 5% read A and C, 4% read B and C, and 2% read all 3 newspapers. A person is selected at random. Use a Venn diagram to help determine the probability that the person reads :

1. none of the papers
2. at least one of the papers
3. exactly one of the papers
4. A or B (or both)
5. A, given that the person reads at least one paper
6. C, given that the person reads either A or B or both.

EXERCISE 44 On any day, the probability that a boy eats his prepared lunch is 0.5. The probability that his sister eats her lunch is 0.6. The probability that the girl eats her lunch given that the boy eats his is 0.9. Determine the probability that :

1. both eat their lunch
2. the boy eats his lunch given that the girl eats hers
3. at least one of them eats their lunch.

c) Independence

Property

A and B are independent events if the occurrence of each one of them does not affect the probability that the other occurs.

This means that $\mathrm{P}(A \mid B) = \mathrm{P}(A \mid B') = \mathrm{P}(A)$.

Property

A and B are independent events $\Leftrightarrow$ $\mathrm{P}(A \cap B) = \mathrm{P}(A)\,\mathrm{P}(B)$

Example

When two coins are tossed, A is the event of getting 2 heads. When a die is rolled, B is the event of getting a 5 or 6. Show that A and B are independent events.

Example

Suppose $\mathrm{P}(A) = \frac{1}{2}$, $\mathrm{P}(B) = \frac{1}{3}$, and $\mathrm{P}(A \cup B) = p$. Find p if :

1. A and B are mutually exclusive
2. A and B are independent.

Example

Suppose $\mathrm{P}(A) = \frac{2}{5}$, $\mathrm{P}(B \mid A) = \frac{1}{3}$, and $\mathrm{P}(B \mid A') = \frac{1}{4}$. Find :

1. $\mathrm{P}(B)$
2. $\mathrm{P}(A \cap B')$

EXERCISE 45 If $\mathrm{P}(X) = 0.5$, $\mathrm{P}(Y) = 0.7$, and X and Y are independent events, determine the probability of the occurrence of :

1. both X and Y
2. X or Y or both
3. neither X nor Y
4. X but not Y
5. X given that Y occurs.

EXERCISE 46

1. Find the probability of getting at least one six when a die is rolled 3 times.
2. Find the smallest n such that $\mathrm{P}(\text{at least one 6 in } n \text{ rolls of a die}) > 99\%$.

EXERCISE 47 A and B are independent events. Prove that A' and B' are also independent events.

EXERCISE 48 Suppose $\mathrm{P}(A \cap B) = 0.1$ and $\mathrm{P}(A \cap B') = 0.4$. Find $\mathrm{P}(A \cup B')$ given that A and B are independent.

EXERCISE 49 Suppose $\mathrm{P}(C) = \frac{9}{20}$, $\mathrm{P}(C \mid D') = \frac{3}{7}$, and $\mathrm{P}(C \mid D) = \frac{6}{13}$.

1. Find :
   (a) $\mathrm{P}(D)$
   (b) $\mathrm{P}(C' \cup D')$
2. Are C and D independent events ? Give a reason for your answer.